Lab 1: FAICT

the Fact AI Checking Tool

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1 Introduction

Internet users are subjected to rising levels of misinformation from online sources and platforms. Misinformation spreads rapidly across social media, news sites, and online communities, a problem that is heavily exacerbated by the rise of AI-generated content. The scale of the issue is vast, affecting everyday web browsing and social media usage. Approximately 64% of U.S. adults report that misinformation causes major confusion (Cruz & Petrino, 2025). In addition to this, while roughly 94% of adults say they check the accuracy of their news sources, not all do so before sharing information with others (Cruz & Petrino, 2025). Consequently, 97% of those surveyed want stricter measures to limit the spread of misinformation (Cruz & Petrino, 2025).

The current process for combating this issue is often slow, confusing, and time-consuming. Manual validation requires extensive online searches and cross-referencing of sources, placing a heavy burden on the user. Existing content verification features offered by specific websites and platforms are limited in their scope and accessibility. These tools frequently suffer from platform exclusivity, popularity bias, and a primary focus on basic text verification. Current AI and chatbot offerings also fail to provide an intuitive, reliable, all-in-one solution. Manual verification is also easily outpaced by the sheer volume of fake content generated by AI tools.

Ultimately, users lack a universal and on-demand method for quickly verifying the information they encounter on the internet, and the burden of manual validation results in many users skipping the process entirely. The required solution must be automated, fast, and capable of providing clear, evidence-based credibility assessments. Therefore, this paper proposes FAICT, the Fact AI Checking Tool, to help fill these gaps and make browsing the web a more reliable experience across all web-based content.

2 Product Description

FAICT is a comprehensive software tool designed to verify information in real time across text, audio, and video formats. The system works by automatically extracting claims from user-provided content and analyzing them against reliable, trusted sources. The primary goal of FAICT is to provide clear credibility assessments accompanied by cited evidence and to detect AI-generated content. By automating and accelerating the verification process, FAICT's core objective is to help internet users quickly identify misinformation, reduce blind trust in unverified claims, and empower them to make better-informed decisions.

2.1 Key Product Features and Capabilities

FAICT's central capability is cross-platform fact-checking: rather than being tied to a single website or social media platform, the tool allows a user to submit a claim from almost any online source and have it checked against a broad pool of trusted references. Every claim that is checked returns a clear credibility rating of true, misleading, false, or uncertain, paired with a short explanation and links to the supporting sources, so a user is never left with just a label and no evidence to evaluate.

Because misinformation increasingly spreads through spoken and recorded content rather than plain text, FAICT extends verification beyond written claims. A built-in speech-to-text module listens to podcasts, livestreams, and real-time conversations and checks claims as they are spoken, and a companion video-processing module converts uploaded video into a transcript so that claims made on camera can be checked the same way. Together, these two capabilities let FAICT treat audio and video as first-class input rather than requiring a user to transcribe them manually first.

Separately from judging whether a claim is true, FAICT also evaluates whether the content itself was produced by AI. Its AI content-detection module screens submitted text, images, audio, and video for signs of generation or heavy modification by AI tools, giving users a second, independent signal alongside the credibility score.

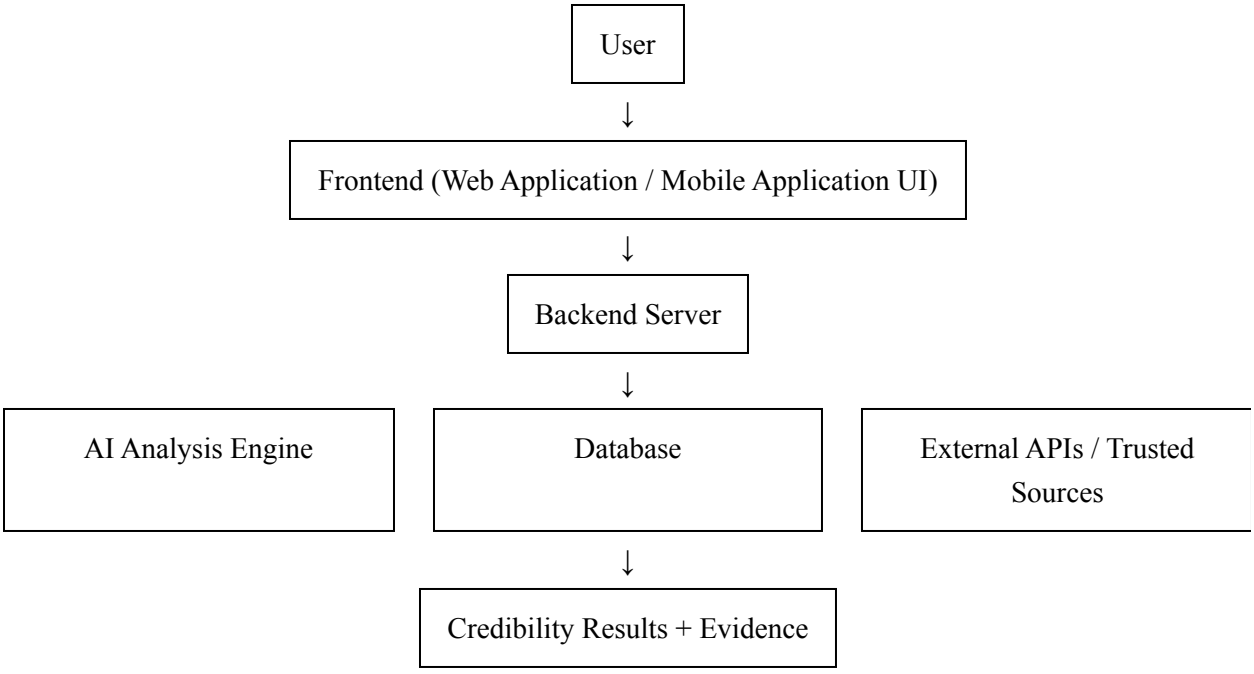
Finally, these capabilities are only useful if a user can reach them at the moment they need them, so FAICT automates the entire fact-checking process end to end and packages it as a browser extension and a mobile application. A user can verify a claim without leaving the page, video, or conversation they are already in, which keeps the tool fast enough to use at the moment misinformation is actually encountered.

2.2 Major Components (Hardware/Software)

FAICT is a software-based solution delivered entirely over the internet rather than through dedicated hardware. It is being built on a Python backend, an HTML/CSS/JavaScript frontend interface, and a PostgreSQL or MongoDB database. Figure 1 lays out how these pieces are organized, and the architecture is described below in the same order a request moves through the system, starting with the user and ending with the evidence returned to them.

Figure 1

Major Functional Components Diagram



On the user side, the frontend interface is the web application and mobile application a person actually interacts with: it accepts a claim in whatever form the user has it – typed text, an audio clip, or an uploaded video – and it is also where the finished credibility results and supporting evidence are displayed once an analysis is complete. This is the only part of the system the user directly sees, so it is responsible for making the input and output of the fact-checking process as simple as possible.

Every request from the frontend passes through the backend server, which acts as the central integration point connecting the frontend to the database, the AI analysis engine, and the external APIs. The AI analysis engine itself handles claim analysis and classification to determine how credible a submitted piece of information is, and it relies on a set of processing

modules – covering speech-to-text conversion, video-to-transcript conversion, and AI-generated content detection – to first turn audio and video input into a form the engine can analyze.

Two supporting components round out the architecture. The database securely stores user data, submission records, structured claims, and analysis results, so a user's history and prior results are not lost between sessions. The external APIs connect the system to outside web searches and trusted fact-checking sources, pulling in the supporting evidence that is ultimately attached to each credibility result before it is displayed back to the user.

3 Identification of Case Study

To justify the need for FAICT and to ground its feature set in observed real-world limitations, this section examines Snopes, one of the most established fact-checking services currently available, as a case study.

Snopes operates as a human-staffed fact-checking publication. When a claim circulates online, members of Snopes' editorial staff select it for review, research it using primary sources, interviews, and public records, and publish a rated article explaining whether the claim is true, false, or a mixture of both (Snopes, n.d.). This approach has made Snopes one of the most recognized and trusted names in fact-checking, and its article archive is frequently cited by journalists, researchers, and casual readers investigating a specific claim.

However, Snopes' manual, editorially-driven model reveals several limitations that directly motivate FAICT's design. First, because every claim must be researched and written up by a human staff member, the service can only cover a small fraction of the claims circulating online at any given time, and claims deemed less viral or less newsworthy may never be reviewed at all. Second, this review process is inherently slow; a new claim can take hours or days to be published, during which time it may continue to spread unchecked. Third, Snopes' coverage is limited almost entirely to text-based claims circulating in articles, memes, and social media posts, with no built-in mechanism for verifying spoken claims in podcasts, interviews, or video content. Finally, as AI-generated misinformation has become easier to produce at scale, Snopes has no dedicated capability for detecting AI-generated content itself, leaving it reliant on the same human-driven research process regardless of whether a claim originates from a person or a generative model. These limitations have made Snopes' long-term sustainability a subject of concern even among its own staff, given how difficult it is for a small editorial team to keep pace with the sheer volume of claims that now require review (Domonoske, 2017).

FAICT directly targets each of these gaps. Where Snopes relies on manual research, FAICT automates claim extraction and verification against trusted sources (Section 2.1), allowing it to process a far greater volume of claims without a proportional increase in staff. Where Snopes is limited to published text, FAICT's speech-to-text and video-processing modules extend verification to audio and video content (Section 2.1). Where Snopes has no mechanism for identifying AI-generated content, FAICT includes a dedicated AI content-detection module (Section 2.1). Taken together, Snopes illustrates both the enduring demand for trustworthy fact-checking and the specific structural limitations that an automated, multi-modal tool like FAICT is positioned to resolve.

4 Glossary

4.1. **AI** – Artificial Intelligence

4.2. **AI-Generated Content** – Text, images, audio, or video created or heavily modified by artificial intelligence tools.

4.3. **API** – Application Programming Interface

4.4. **Browser Extension** – A small software add-on that adds features or functionality to a web browser.

4.5. **Credibility Score** – A rating that estimates how trustworthy or reliable a claim appears based on supporting evidence.

4.6. **Disinformation** – False or misleading information that is intentionally created or shared to deceive people.

4.7. **Fact-Checking** – The process of verifying whether a claim is accurate by comparing it against reliable sources.

4.8. **FAICT** – the Fact AI Checking Tool

4.9. **GUI** – Graphical User Interface

4.10. **IDE** – Integrated Development Environment

4.11. **LLM** – Large Language Model

4.12. **Misinformation** – False or misleading information that may be shared regardless of whether there is intent to deceive.

4.13. **Speech-to-Text** – Technology that converts spoken audio into written text.

4.14. **Trusted Sources** – Reliable references, databases, publications, or organizations used to verify claims.

4.15. **Verification** – The process of confirming whether information is accurate, reliable, or supported by evidence.

4.16. **Video-to-Transcript** – The process of converting spoken content from a video into written text for analysis.

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